



Determine if each problem when converted to a decimal will result in a repeating (R) or terminating (T) decimal.

Answers

A fraction will result in a **terminating** decimal if the prime factors of the simplified denominator contain only 2s or 5s (or only 2s and 5s).

$$6/40 = 3/20 = 2 \times 2 \times 5 = 0.15$$

A fraction will result in a **repeating** decimal if the prime factors of the simplified denominator contain any prime factor other than 2 or 5.

$$5/42 = 2 \times 3 \times 7 = 0.1190476$$

1) $2/3 =$ _____

2) $19 \div 8 =$ _____

3) $86 \div 20 =$ _____

4) $102 \div 19 =$ _____

5) $135 \div 17 =$ _____

6) $5/27 =$ _____

7) $11/13 =$ _____

8) $164 \div 16 =$ _____

9) $11/30 =$ _____

10) $79 \div 24 =$ _____

11) $2/15 =$ _____

12) $243 \div 26 =$ _____

13) $3/18 =$ _____

14) $8/22 =$ _____

15) $144 \div 14 =$ _____

1. _____

2. _____

3. _____

4. _____

5. _____

6. _____

7. _____

8. _____

9. _____

10. _____

11. _____

12. _____

13. _____

14. _____

15. _____